

TravelEase Hub

Travel Assistance with Pooling System

BINZIYA P

INT MCA S9

TravelEase Hub

ABSTRACT

Introduction

TravelEase Hub is a full-stack web application designed to simplify and personalize travel planning. The platform serves three types of users—travelers, travel agencies, and administrators—and offers features such as AI-powered itinerary planning, smart package search, booking management, budget estimation, and upcoming tools like user pooling and automatic expense splitting. With interactive dashboards and real-time updates, the platform aims to provide a smarter, more collaborative, and highly personalized travel experience.

Existing System

Popular travel platforms like MakeMyTrip, Yatra, Expedia, and Booking.com typically offer static package listings with limited personalization, minimal automation, and no real-time collaborative features. Users often face challenges in planning group travel, estimating shared costs, and customizing itineraries based on specific preferences. The absence of features like budget estimation, expense splitting, and user pooling limits user flexibility and satisfaction.

Proposed System

The proposed TravelEase Hub platform allows users to register, search for travel packages using smart filters, and receive AI-generated itinerary suggestions based on personal preferences. Users can book and cancel trips, estimate budgets, view travel routes on maps, and access trip history through a personalized dashboard. Travel agencies can sign up to list and manage travel packages, create promotional offers, and receive real-time booking alerts. Administrators are responsible for monitoring platform activity, managing user and agency accounts, and verifying the authenticity of submitted packages. Future extensions of the system will include a user pooling feature that matches travelers with similar destinations and a smart expense splitting tool to automatically divide trip costs among group travelers.

Project Features

The system consists of 3 main modules:

1. User Module

- Login/Register: Secure login and registration for users
- Package Search: Filter by destination, budget, and duration
- Booking Management: Book, cancel, and view package details
- AI Itinerary Planner: Generates personalized travel plans
- Budget Estimator: Calculates estimated costs for selected trips
- Map View: Displays travel routes and destinations
- History & Reviews: View past trips and submit feedback

2. Agency Module

- Login/Register: Agencies can securely sign up and log in
- Manage Packages: Add, edit, and delete travel packages
- Booking Alerts: Get real-time notifications for new bookings
- Promotions: Create and manage custom offers and discounts

3. Admin Module

- User & Agency Management: Activate/deactivate user and agency accounts
- Verify Listings: Approve submitted travel packages
- Notifications: Send platform-wide alerts and updates

Upcoming Features (Main Project Extension)

- Pooling Option: Match users traveling to similar places and share travel
- Smart Expense Splitting: Automatically split and manage trip costs among group travelers

Technologies Used

- Frontend: React.js
- Backend: Node.js with Express.js
- Database: Supabase (PostgreSQL-based backend-as-a-service)
- AI/ML: TensorFlow.js / OpenAI API
- APIs: Google Maps API
- Authentication: JWT (JSON Web Token)